



Lecture 2

Basic Laws & Circuit Analysis

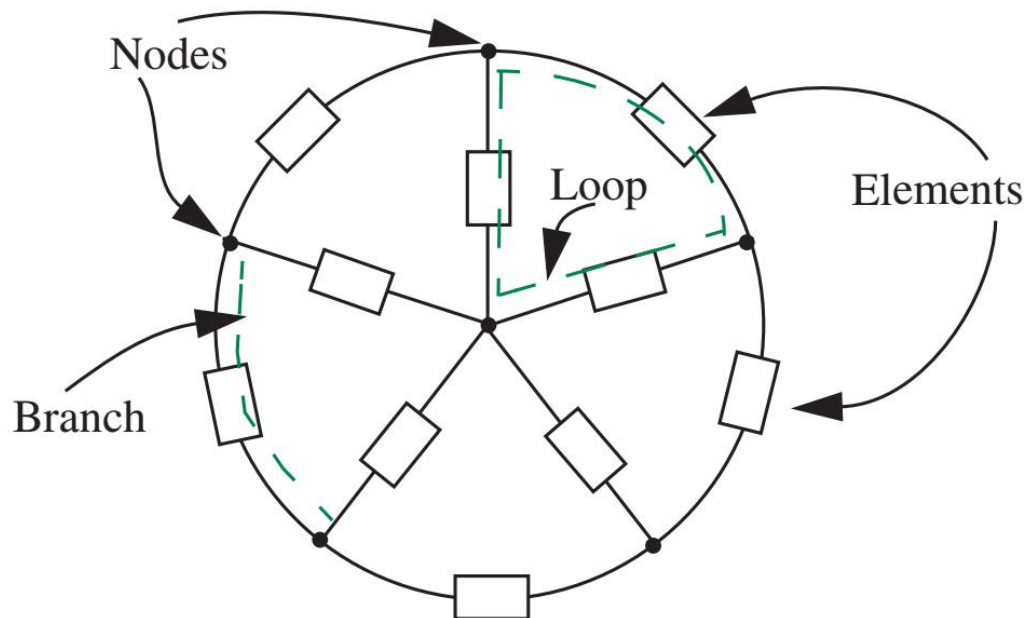


Outline

- Terminology: Branches, Nodes, and Loops
- Kirchhoff's Laws
 - KCL
 - KVL
- Circuit Analysis
 - Nodal Analysis
 - Mesh Analysis

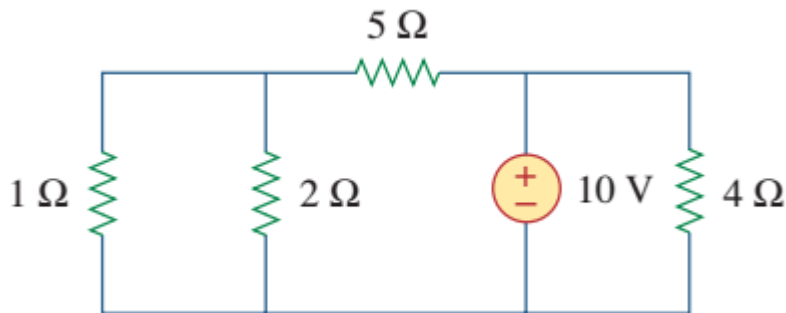
Terminology: Branches, Nodes, and Loops

- **Branch**: represents a single element;
- **Node**: a point of connection between two or more branches;
- **Loop**: **any** closed path in a circuit.



Loop, Independent Loop, Mesh

- A loop is a closed path with no node passed more than once.
- A loop is independent if it contains at least one branch which is not a part of any other independent loop.
- A mesh is a loop that does not contain any other loop within it.



- b – number of branches
- n – number of nodes
- l_{ind} – number of ind. loops

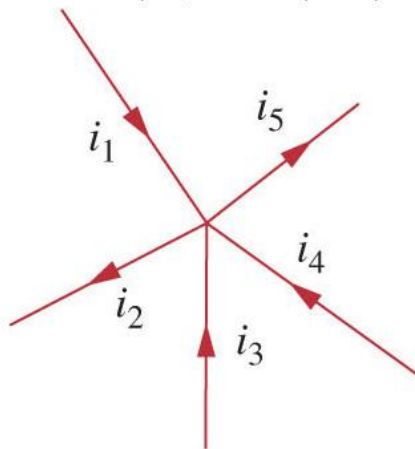
$$l_{ind} = b - (n - 1)$$



Kirchhoff's Laws

- Kirchhoff's Current Law (KCL):
 - The algebraic sum of all the **currents** entering any **node** in a circuit equals zero.
 - **Why?**

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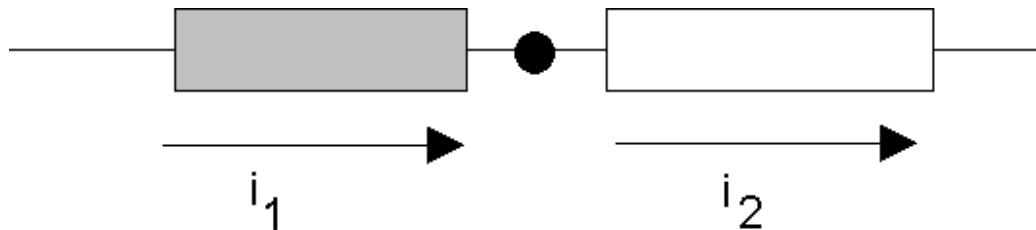


Gustav Robert Kirchhoff
1824-1887



A Major Implication of KCL

- KCL tells us that **all** of the elements that are connected ***in series*** carry the same current.

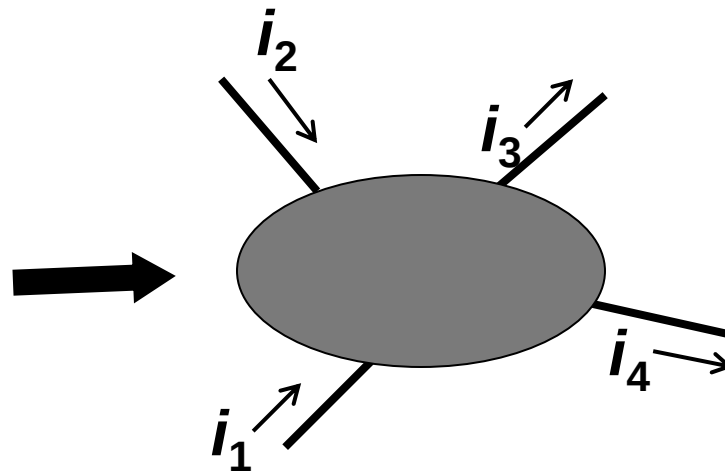


Current entering node = Current leaving node

Generalization of KCL

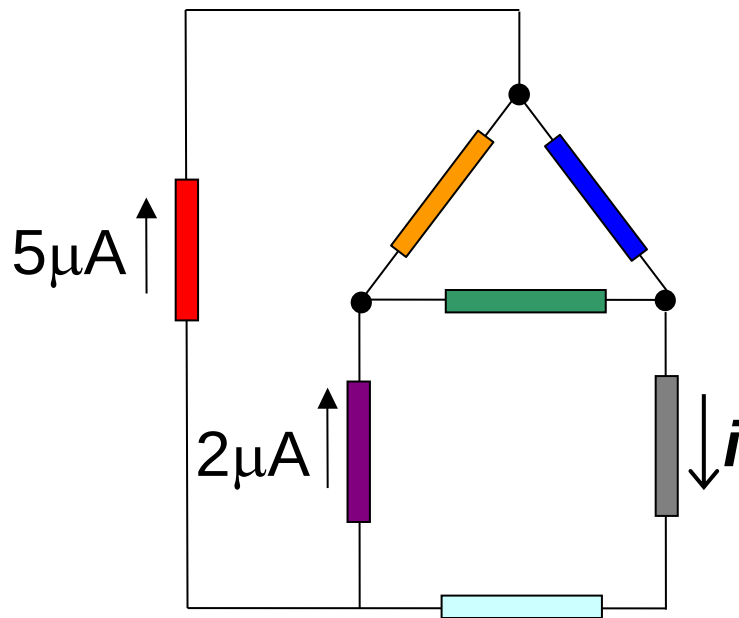
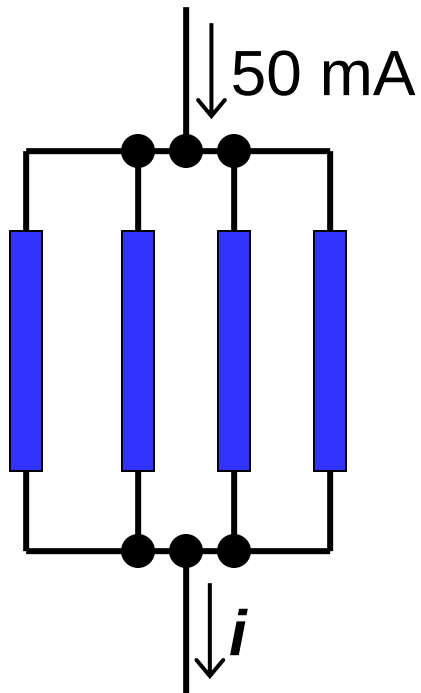
- The sum of currents entering/leaving a **closed surface** is zero.
 - Circuit branches can be inside this surface, *i.e.* the surface can enclose more than one node!

This could be a big chunk of a circuit, *e.g.* a “black box”

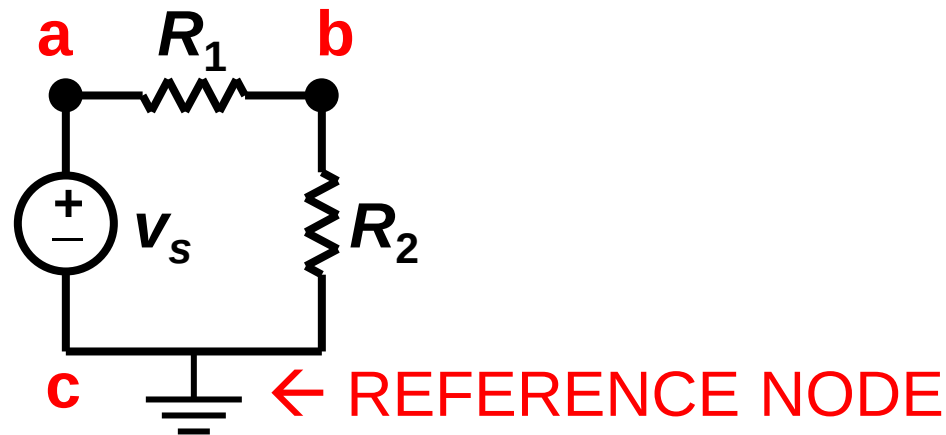




Generalized KCL Examples



Notation: Node and Branch Voltages



- Use one node as the reference (the “common” or “ground” node) – label it with a symbol.
- The voltage drop from node **x** to the reference node is called the **node voltage** V_x .
- The voltage across a circuit element is defined as the difference between the node voltages at its terminals.

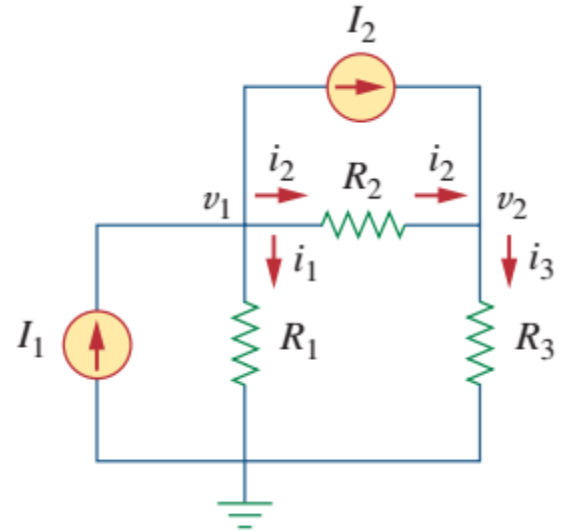
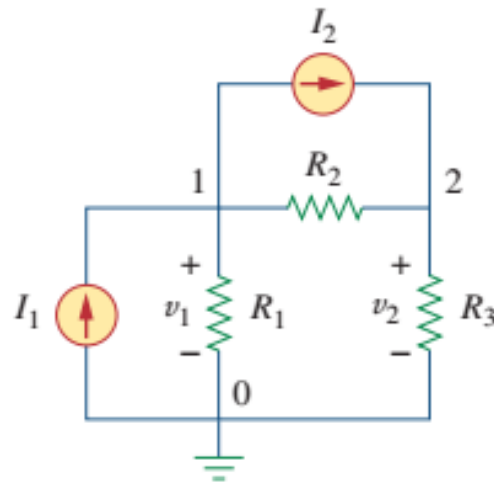
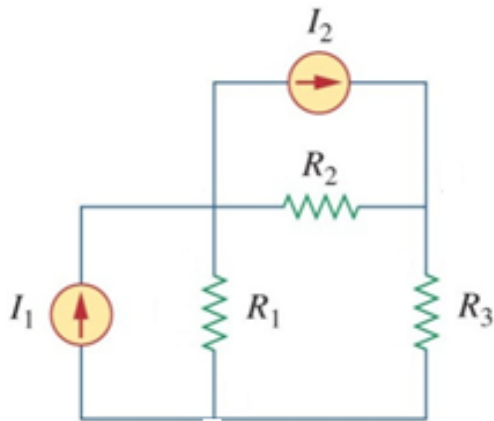


Nodal Analysis – Three Steps

- Given a circuit with n nodes, the nodal analysis is accomplished via three steps:
 1. Select a node as the reference (i.e., ground) node. Assign the node voltages to the remaining **$(n-1)$** nodes. Voltages are relative to the reference node.
 2. Apply KCL to the **$(n-1)$** nodes, expressing branch current in terms of the node voltages (using the I - V relationships of branch elements).
 3. Solve the resulting simultaneous equations to obtain the unknown node voltages.

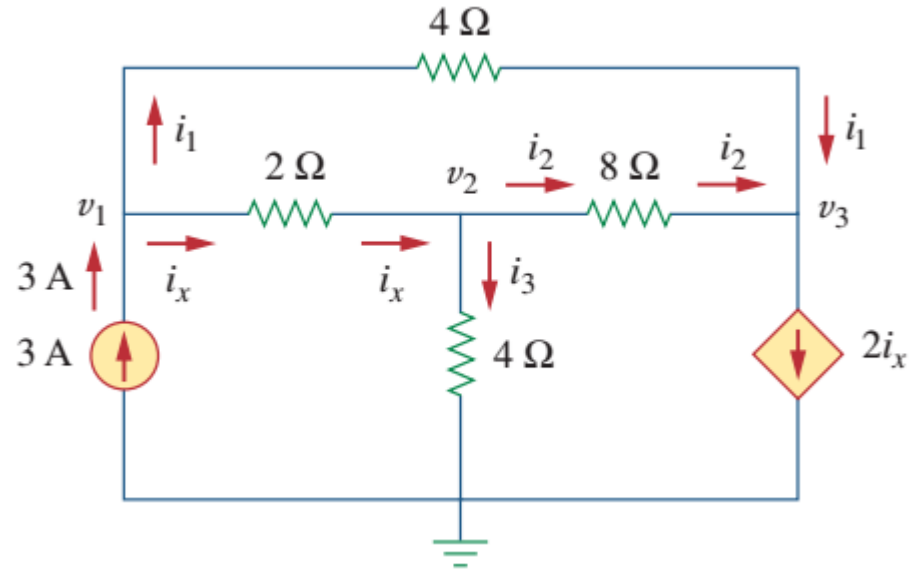
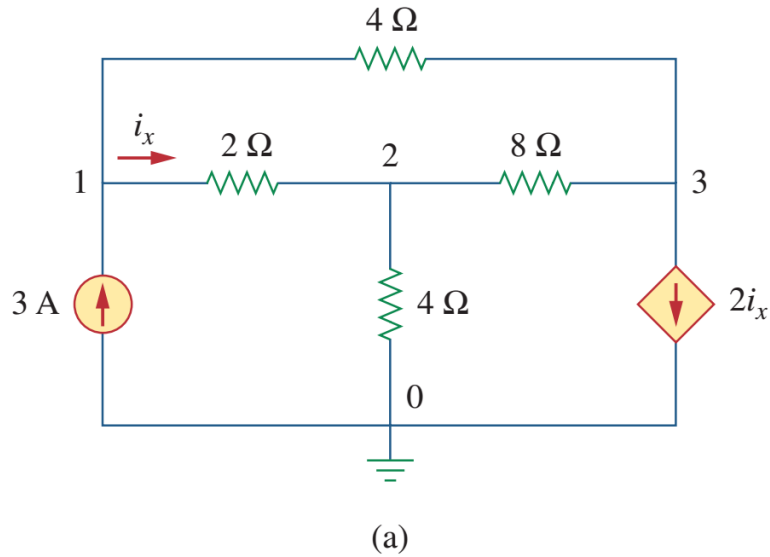


Nodal Analysis Example #1





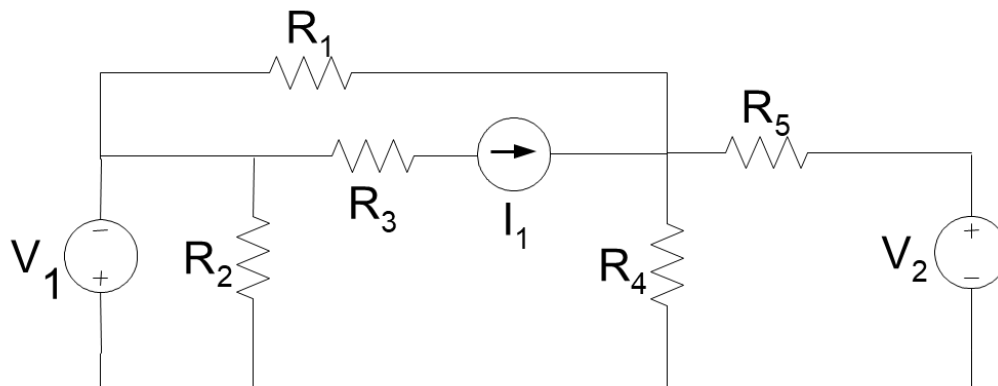
Nodal Analysis: Example #2





Nodal Analysis with Voltage Sources

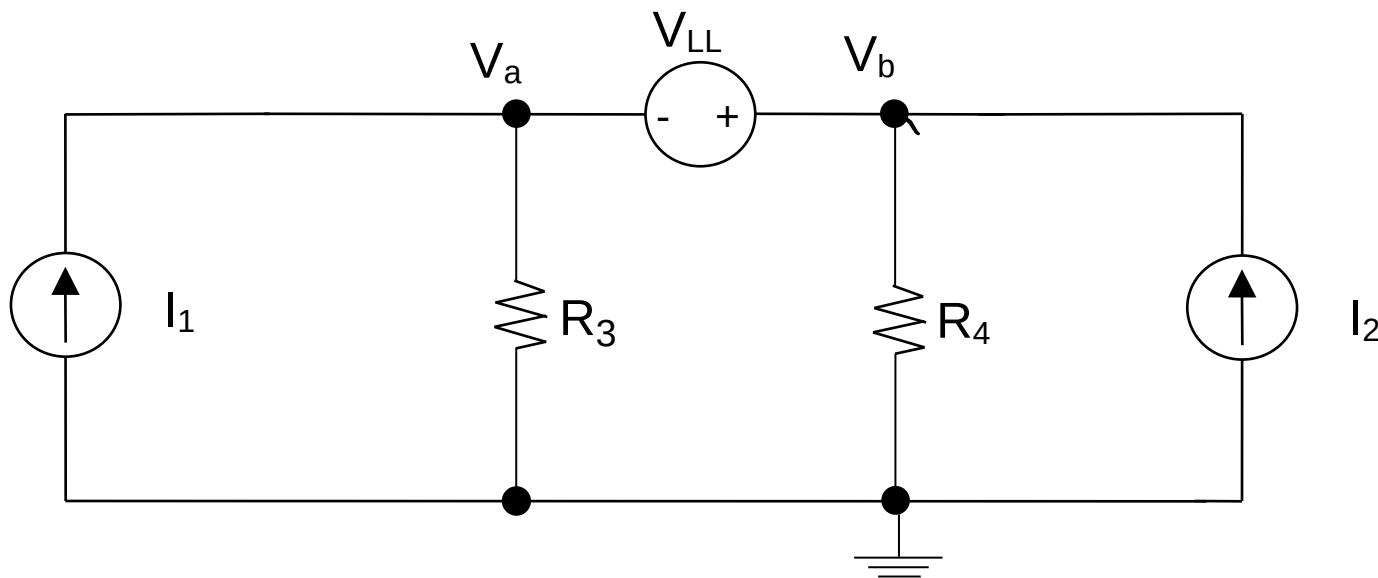
Case I:



Nodal Analysis: Supernode

Case II

A “floating” voltage source is one for which **neither** side is connected to the reference node, e.g. V_{LL} in the circuit below:

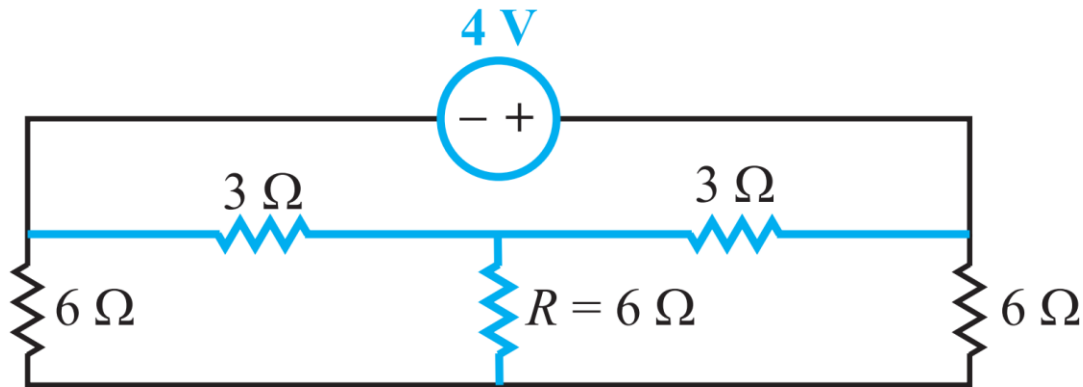


A supernode is formed by enclosing a (dependent or independent) voltage source connected between two nonreference nodes and any elements connected in parallel with it.



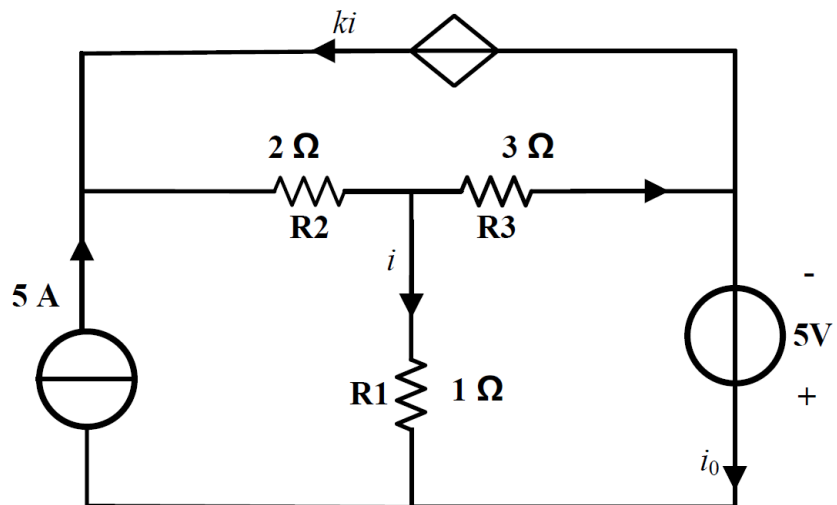
Exercise

- Find the power supplied by the voltage source.





(10 points) The circuit is shown in Figure 1. Known that $k = 2$. Using nodal analysis to find i_0 .

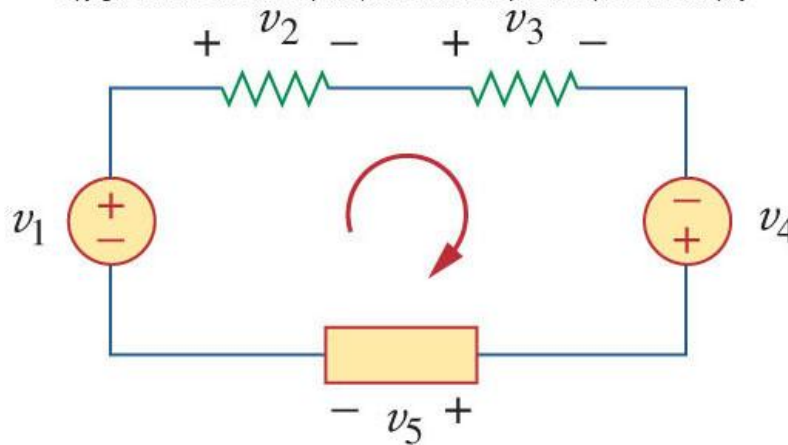




Kirchhoff's Voltage Law (KVL)

- The algebraic sum of all the **voltages** around any **loop** in a circuit equals zero.
- **Why?**

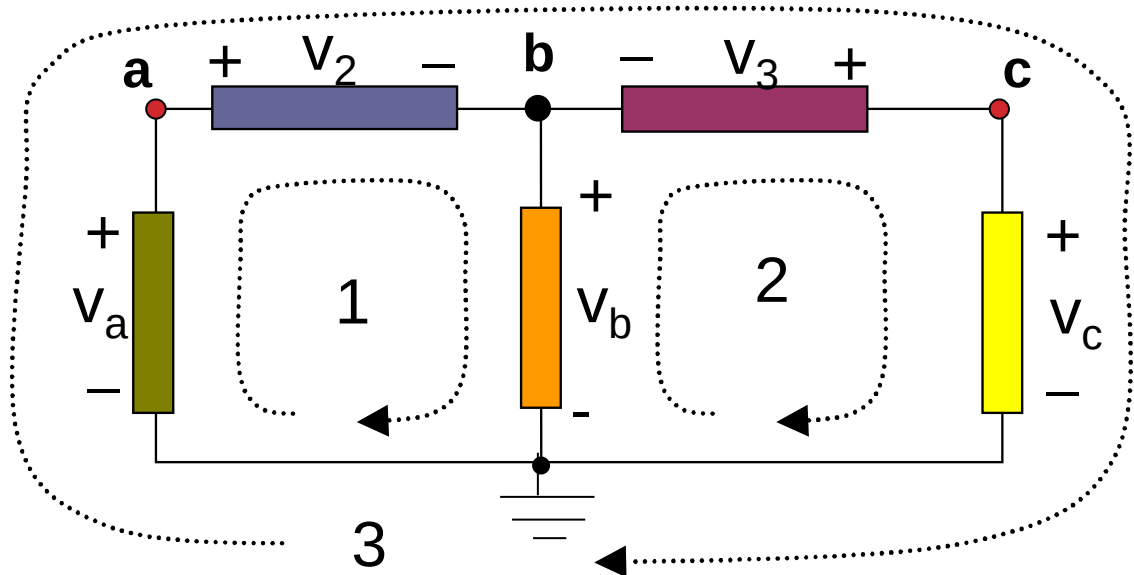
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KVL Example

Three closed paths:



Path 1:

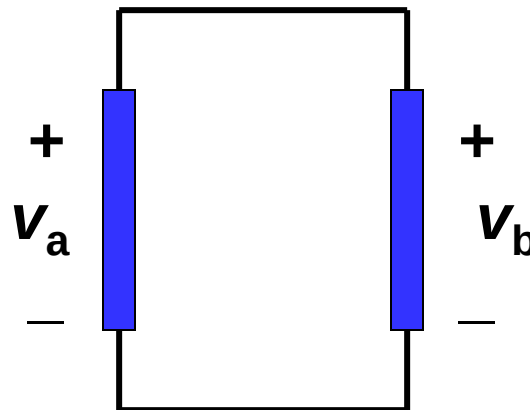
Path 2:

Path 3:



A Major Implication of KVL

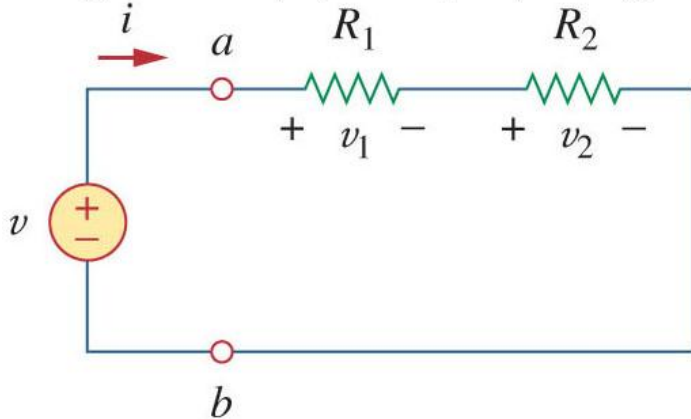
- KVL tells us that **any set of elements which are connected at both ends carry the same voltage.**
- We say these elements are connected **in parallel**.



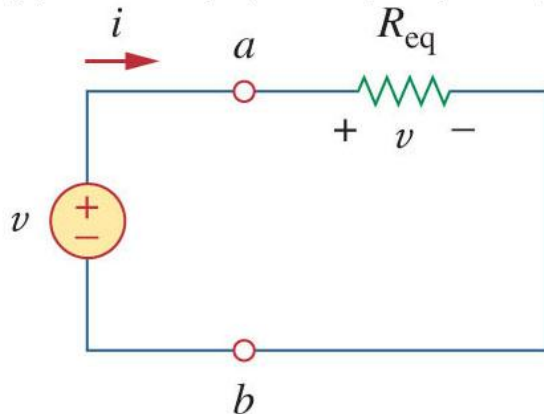


Voltage Division

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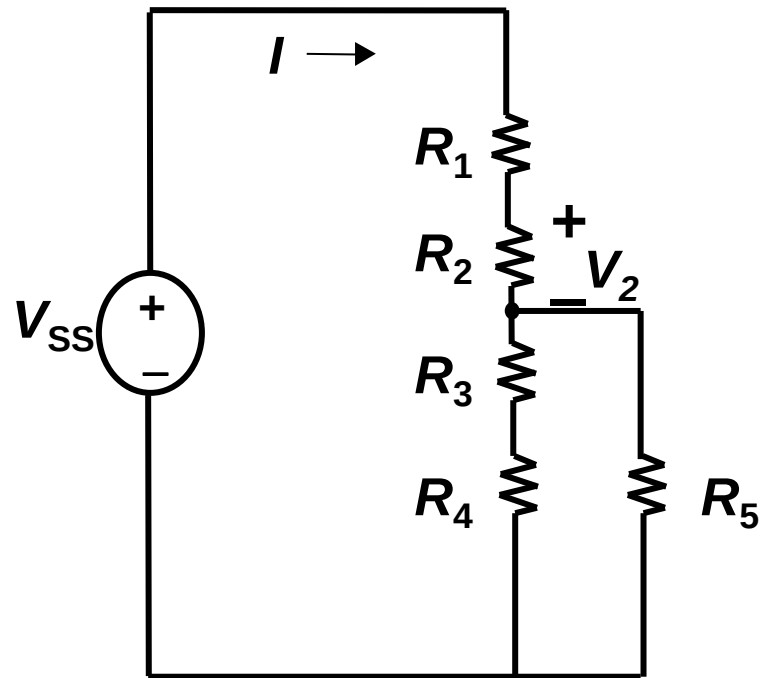
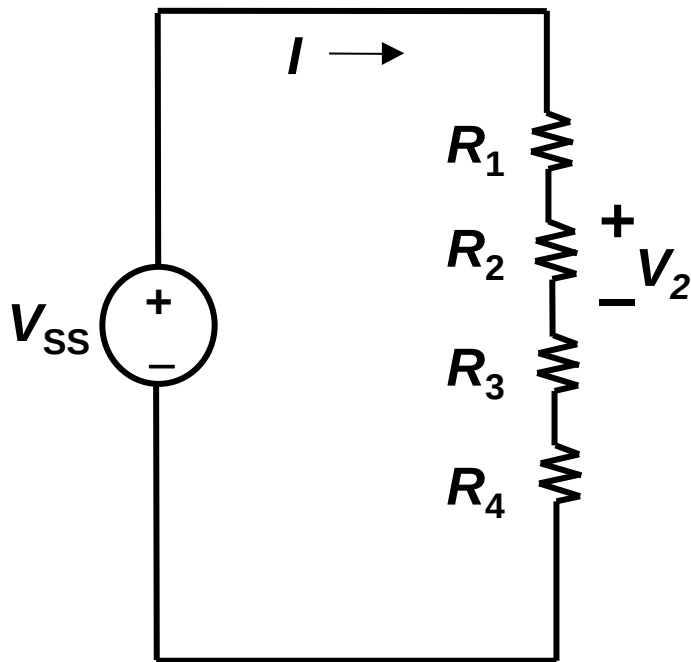
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Three-terminal rheostat



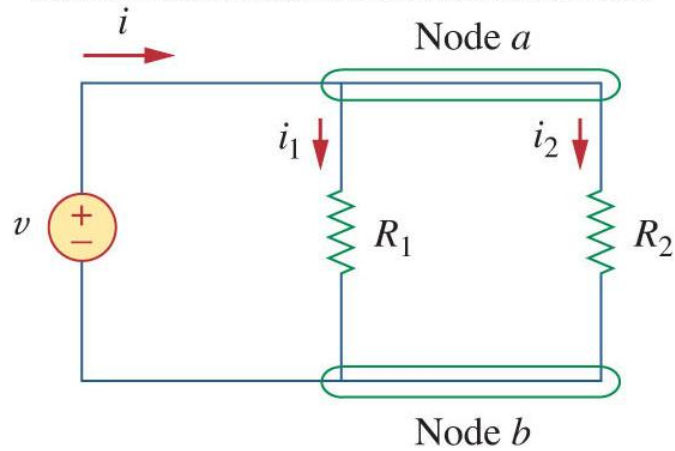
Voltage Divider



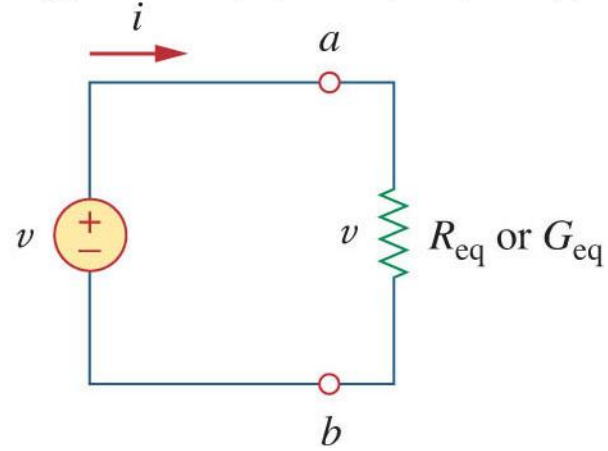


Parallel Resistors/Current Division

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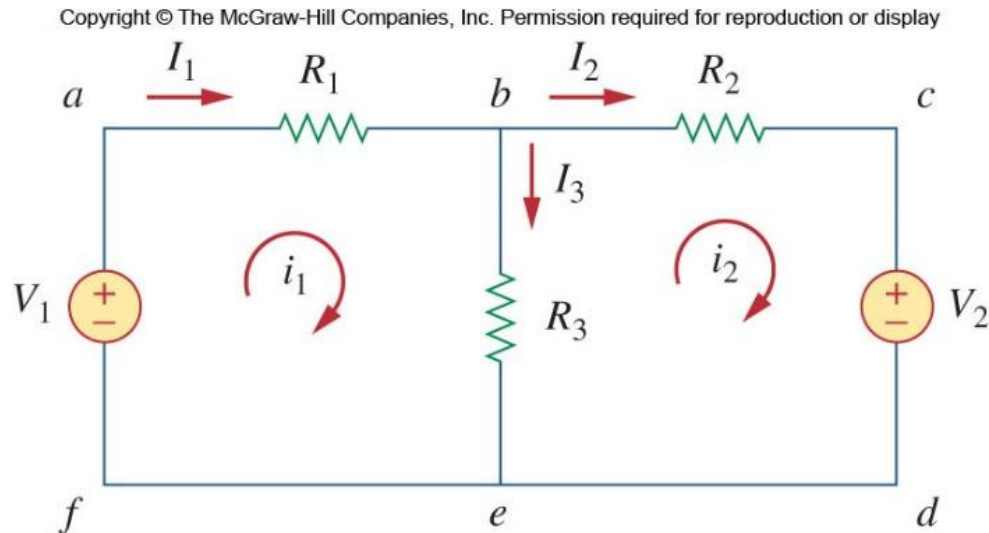
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Mesh Analysis

- Another general procedure for analyzing circuits is to use the mesh currents as the circuit variables.

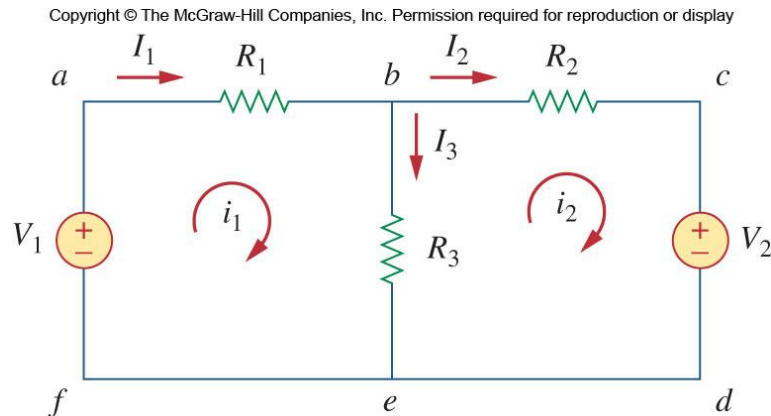


- Mesh analysis uses KVL to find unknown currents.



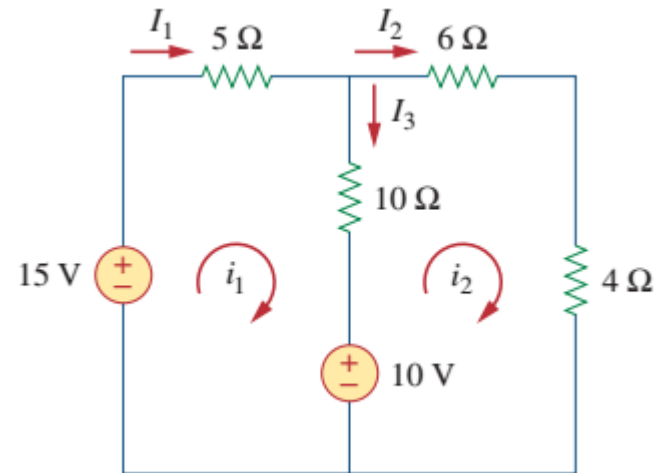
Mesh Analysis Steps

- Mesh analysis follows these steps:
 1. Assign mesh currents $i_1, i_2, \dots, i_x$ to the x meshes
 2. Apply KVL to each of the x mesh currents.
 3. Solve the resulting x simultaneous equations to get the mesh currents.



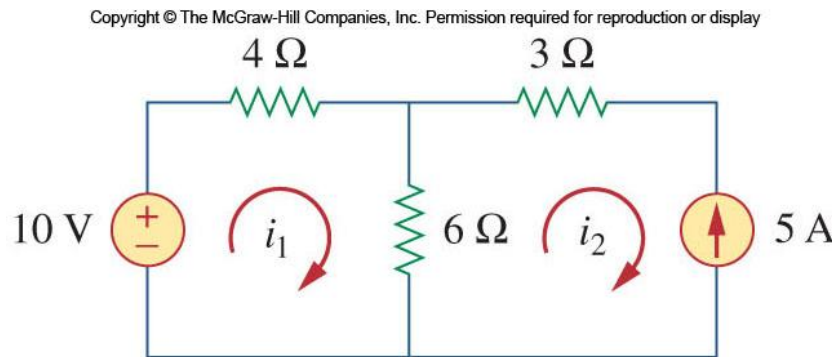


Example



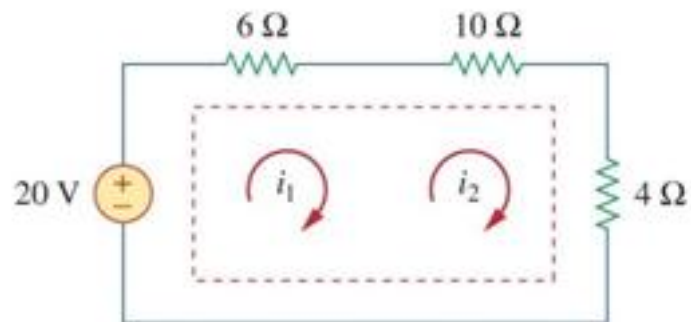
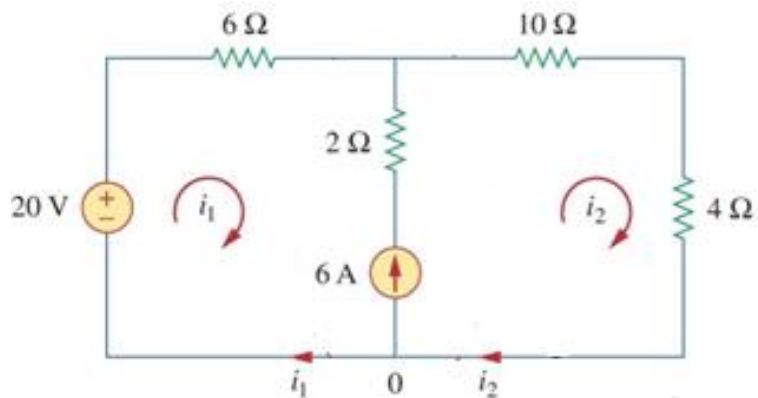
Mesh Analysis with Current Sources

- The presence of a current source makes the mesh analysis simpler in that it reduces the number of equations.
 - If the current source is located on only one mesh, the current for that mesh is defined by the source. For example:





Supermesh



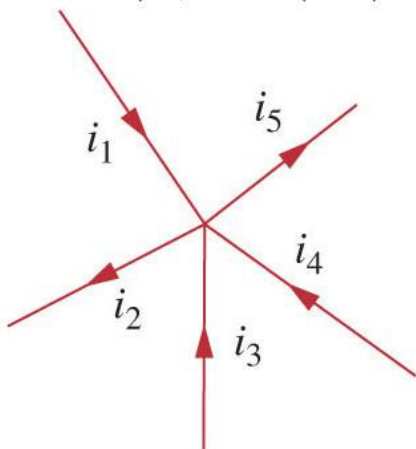
Summary

- KCL and KVL

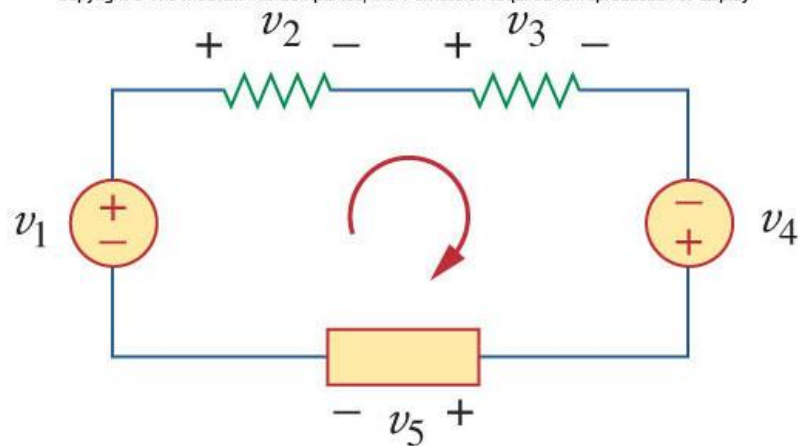
$$\sum_{n=1}^N i_n = 0$$

$$\sum_{m=1}^M v_m = 0$$

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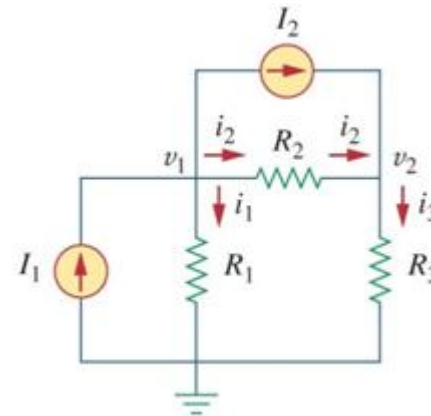


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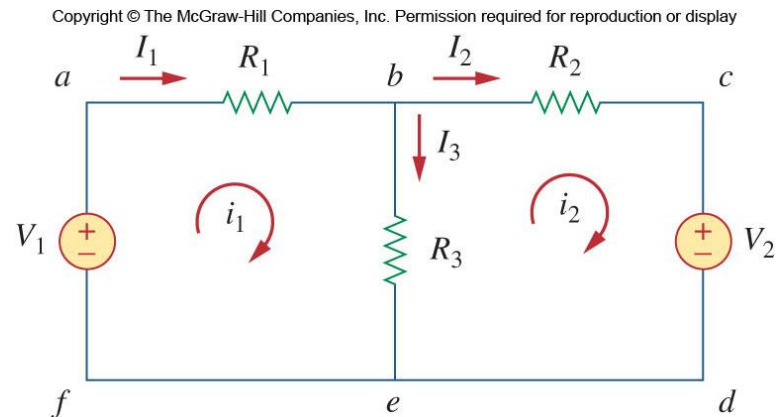


Summary

- Node Analysis
 - Node voltage is the unknown
 - Solve by KCL
 - Special case: Floating voltage source

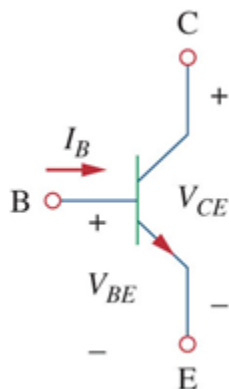


- Mesh Analysis
 - Loop current is the unknown
 - Solve by KVL
 - Special case: Current source

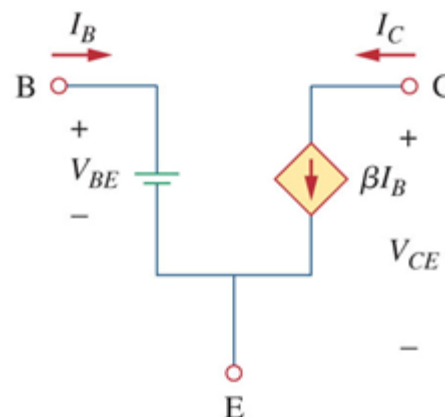


DC model of a BJT

- The figure below shows the equivalent DC model for a BJT in active mode.



$$I_C = \beta I_B$$

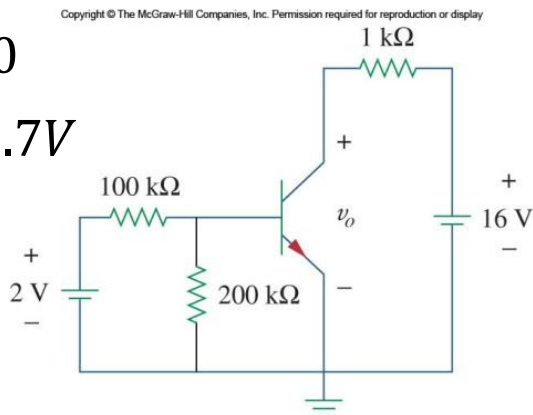




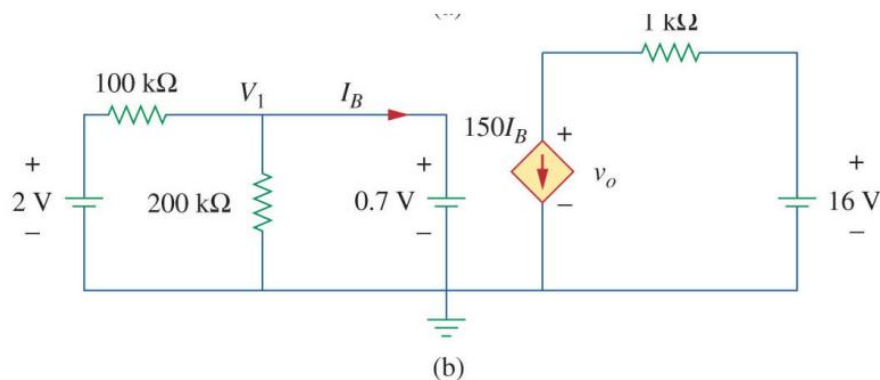
Setting up a BJT circuit

$$\beta = 150$$

$$v_{BE} = 0.7V$$



Original circuit



Circuit for nodal analysis



Wye-Delta Transformations